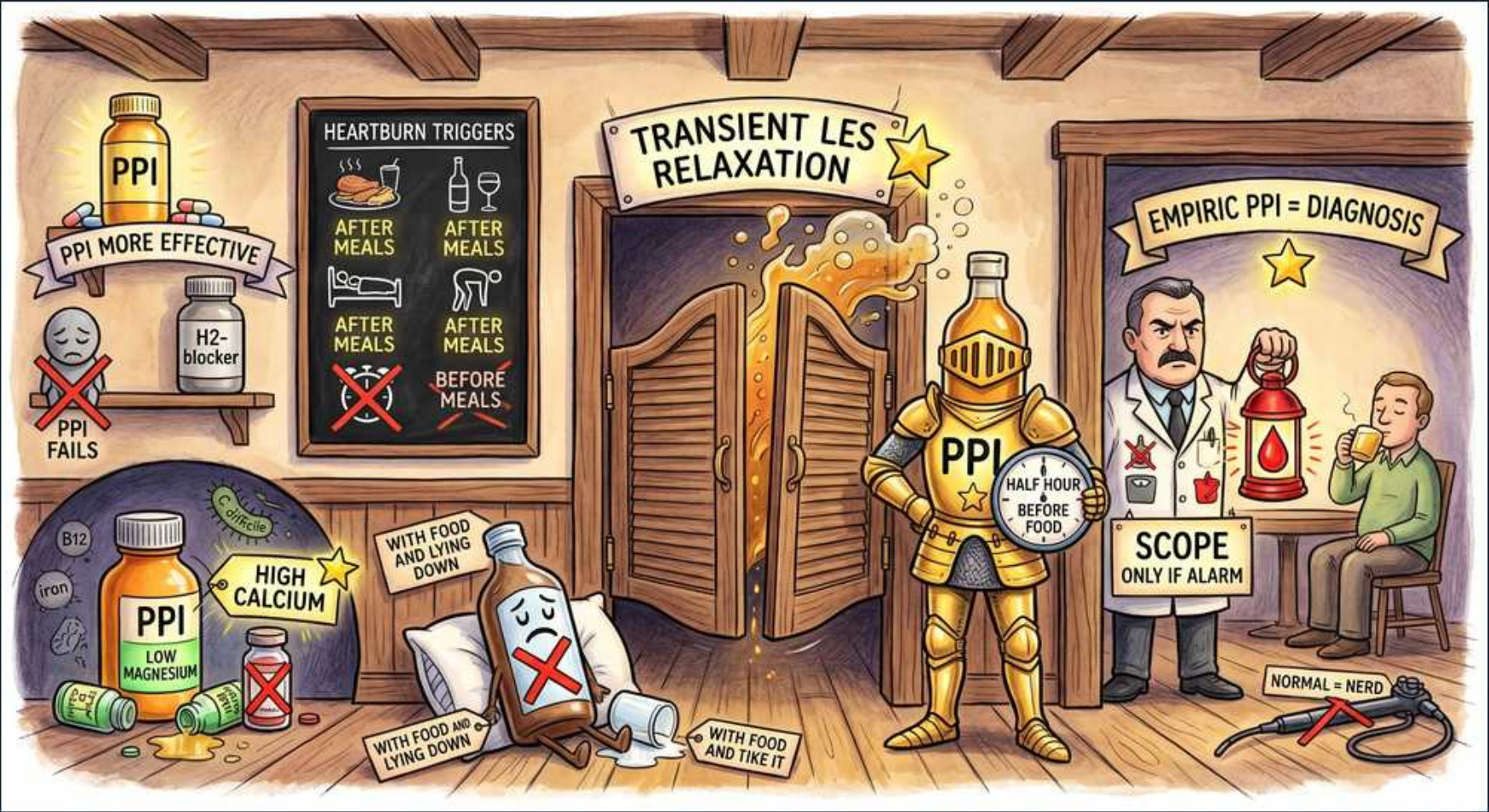


Esophageal — GERD



1. Transient LES relaxation

WHAT YOU SEE

Saloon doors swinging open with reflux spraying over them, banner 'TRANSIENT LES RELAXATION'.

WHAT IT ENCODES

The dominant pathophysiology of GERD is transient lower-esophageal-sphincter relaxations (tLESRs), not a chronically weak sphincter alone. Hiatal hernia, obesity and delayed gastric emptying contribute. Acid (and sometimes bile) refluxes into the esophagus.

2. Empiric PPI = diagnosis (gold star)

WHAT YOU SEE

Gold-star banner 'EMPIRIC PPI = DIAGNOSIS' over the bartender.

WHAT IT ENCODES

GERD is a CLINICAL diagnosis; typical heartburn/regurgitation without alarm features warrants an empiric PPI trial (once daily x8 weeks). Symptom response confirms it. No scope or pH study needed for the classic uncomplicated case.

3. Scope only if alarm

WHAT YOU SEE

Banner 'SCOPE ONLY IF ALARM' below the bartender.

WHAT IT ENCODES

Endoscopy is reserved for ALARM features: dysphagia, odynophagia, weight loss, GI bleeding/anemia, recurrent vomiting, or new symptoms at older age (and to screen for Barrett's in chronic GERD with risk factors). Routine scoping of typical reflux is not indicated.

4. Scope everyone (RED X concept)

WHAT YOU SEE

Small endoscope with a red X at lower right.

WHAT IT ENCODES

Endoscoping every reflux patient is wrong; classic GERD is diagnosed and treated empirically. Scope only for alarm features or Barrett's surveillance. Over-scoping is low yield and costly.

BOARD TRAP

'endoscopy for everyone with reflux' -> wrong; scope only with alarm features; otherwise empiric PPI.

5. PPI more effective than H2

WHAT YOU SEE

PPI box on a shelf with banner 'PPI MORE EFFECTIVE'.

WHAT IT ENCODES

PPIs heal erosive esophagitis and control symptoms better than H2-blockers (which also lose potency via tachyphylaxis). PPI is first-line maintenance for erosive/severe disease. H2-blockers are for mild/intermittent symptoms or nocturnal breakthrough.

6. H2-blocker / PPI fails

WHAT YOU SEE

'H2-BLOCKER' bottle with a sad face and 'PPI FAILS' red X below.

WHAT IT ENCODES

If a PPI 'fails', first verify adherence and timing (30-60 min before meals) and twice-daily dosing before escalating. H2-blocker is weaker than PPI and not the upgrade. Refractory true GERD -> pH-impedance testing, consider fundoplication.

7. Take PPI 30 min before food

WHAT YOU SEE

Knight's belt buckle reads 'HALF HOUR BEFORE FOOD'.

WHAT IT ENCODES

PPIs are prodrugs that must covalently bind ACTIVATED proton pumps; take ~30-60 min BEFORE a meal so peak drug level coincides with meal-stimulated pump activity. Taking it with food or at bedtime fasting blunts efficacy.

8. With food / take it (RED X)

WHAT YOU SEE

Bottle on a pillow with red X 'WITH FOOD AND TAKE IT'.

WHAT IT ENCODES

Taking a PPI WITH food is wrong: fewer pumps are in the activatable state when the drug peaks, reducing acid suppression. Correct timing is 30-60 min before the meal.

BOARD TRAP

'take PPI with food' -> wrong; take 30-60 min before meals so it inhibits meal-activated proton pumps.

9. With food and lying down (RED X)

WHAT YOU SEE

Bottle reclining on a pillow, red X 'WITH FOOD AND LYING DOWN'.

WHAT IT ENCODES

Lying down after eating worsens reflux (gravity no longer clears acid) and mistimes the PPI. Lifestyle advice: don't recline within 3h of meals, elevate the head of bed, weight loss, avoid late meals.

BOARD TRAP

'take PPI with food and lie down' -> wrong; recumbency worsens reflux and timing is off; dose before meals, stay upright.

10. Low magnesium (PPI side effect)

WHAT YOU SEE

Orange bottle labeled 'LOW MAGNESIUM' with B12/Ca tablets spilled.

WHAT IT ENCODES

Long-term PPI side effects: hypomagnesemia, plus impaired absorption of B12, iron and calcium (-> fracture risk), increased C. difficile and community pneumonia, possible CKD/AIN. These reflect reduced gastric acidity.

11. High calcium (RED X)

WHAT YOU SEE

Tablet labeled 'HIGH CALCIUM' with a red X.

WHAT IT ENCODES

Hypercalcemia is NOT a PPI effect; PPIs actually impair calcium absorption (favoring low Ca and fractures). The classic electrolyte association is hypoMAGNESemia. High calcium on a stem points elsewhere.

BOARD TRAP

'hypercalcemia as a PPI side effect' -> wrong; PPIs cause hypomagnesemia and reduce Ca absorption, not raise it.

12. Heartburn worse after meals/lying

WHAT YOU SEE

Chalkboard 'HEARTBURN TRIGGERS: AFTER MEALS x4 / BEFORE MEALS (X)'.

WHAT IT ENCODES

Reflux heartburn classically worsens AFTER meals and when lying down or bending (postprandial gastric distension + recumbency). 'Before meals / relieved by food' pattern suggests duodenal ulcer instead, a useful discriminator.

13. NERD = normal endoscopy

WHAT YOU SEE

Label 'NORMAL = NERD' with a scope at lower right.

WHAT IT ENCODES

Non-erosive reflux disease (NERD) = typical reflux symptoms with a NORMAL endoscopy (no erosions/Barrett's). Still real GERD; confirm with pH-impedance if needed. Manage with PPI/lifestyle as in erosive disease.